

JACOB ARAGAO

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WORK EXPERIENCE

NationGraph

Software Engineer Intern | Go, TypeScript, React, Python, PostgreSQL, Tailwind

05/2026 - 08/2026

- Architected an end-to-end AI recommendations engine in Go and React, orchestrating 3 parallel streamed LLM calls to reduce round-trip processing latency by ~50%.
- Profiled and optimized high-volume database queries using EXPLAIN ANALYZE, slashing 504-prone response times from ~10s to ~30ms and stabilizing production for 10k-record accounts.
- Proactively identified and remediated an IDOR authentication vulnerability by refactoring user routes across FastAPI and UI services to derive identity directly from signed JWT claims.
- Built 16 accessible, tested UI components comprising ~26% of the shared design system and migrated the primary web application to Tailwind CSS v4 across 110+ merged pull requests.

CareHive Health

Software Engineering Intern | TypeScript, SolidJS, Node.js, PostgreSQL, Vitest

05/2025 - 08/2025

- Built and shipped a full-stack **Reminder Engine** to re-engage members; scaled architecture from **0 to 9 services** with per-service checkpoints and deep links.
- Designed a custom **sorting algorithm** to rank and surface the highest-value user actions based on dynamic priority metrics.
- Refactored logic into isolated components and added extensive **unit tests (Vitest)**, improving reliability and developer velocity.

Clearer.io

Junior Software Engineer | React, Node.js, PostgreSQL, Redis, Stripe

09/2024 - 04/2025

- Built data ingestion workflows using **Node.js** and **webhooks** to sync real-time tracking data from ShipStation, increasing shipment visibility by **50%**.
- Independently designed and deployed "Instant Exchanges," integrating **Stripe** payments directly into the return flow; streamlined exchanges for **39%** of users.

Software Engineering Intern | Node.js, AWS S3, Sequelize ORM

05/2024 - 08/2024

- Executed critical **PostgreSQL** database migrations using **Sequelize ORM**, resolving a production issue affecting over **900 users** with minimal downtime.
- Developed a backend API endpoint in **Node.js** to securely remove AWS S3 photo references, ensuring complete data consistency across the database and cloud storage.

EDUCATION

McGill University, Montreal, Quebec

08/2022 - 04/2027 (Expected)

Bachelor of Engineering in Software Engineering (Co-op)

- **Academic Distinction:** Placed in the top 15% of the Faculty of Engineering (2024-2025).

PROJECTS

Vantage

youtu.be/O69Y7m8nO_I

- Architected a real-time multimodal voice assistant using **Django Channels** and **Gemini 2.5 Flash API**, implementing full-duplex **WebSockets** for simultaneous audio capture and screen context injection.
- Containerized the application with **Docker Compose**, orchestrating the Daphne ASGI server and MongoDB.

FantasyGuide

youtu.be/EM8tWZOGKWI

- Engineered a GenAI data pipeline in **Python** that ingests raw player data via **BeautifulSoup**, cleanses it using **Pandas**, and feeds it into the **OpenAI API**.
- Developed the frontend interface using **React** and **Flask**, integrating AI recommendations to significantly enhance user engagement.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Go, SQL, GraphQL, HTML, CSS, Java, C
- **AI & Data:** OpenAI API, Prompt Engineering, RAG, Scikit-learn, Pandas, NumPy
- **Full Stack:** React, SolidJS, Tailwind CSS, Node.js, Express, Django (Channels), Flask, WebSockets
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, AWS (S3, CloudWatch), Docker
- **DevOps & Testing:** Git, GitHub, GitLab CI/CD, Jira, Vitest, Jest, pytest-asyncio